

Lab 2

Problem 1:

- Run a container nginx with name my-nginx and attach 2 volumes to the container using volume mount
- Volume1 for containing static html file
- Volume2 for containing nginx configuration
- Edit the html content
- Remove the container
- Run a new 2 containers with the following:
 - o Attach the two volumes that were attached to the previous container using volume mount
 - o Map port 80 to port 8080 on you host machine
 - o Access the html files from your browser

Terminal

```
PS C:\Users\Marawan Ragab\my_web_project> docker volume create vol-html
vol-html
PS C:\Users\Marawan Ragab\my_web_project> docker volume create vol-conf
vol-conf
PS C:\Users\Marawan Ragab\my_web_project> 
```

```
PS C:\Users\Marawan Ragab\my_web_project> docker run -d --name my-nginx `
>> -v vol_html:/usr/share/nginx/html `
>> -v vol_config:/etc/nginx/conf.d `
>> nginx
76e2ad484c0c9063bd023eedf8eaf7bf2b33c8e2350de76c1af1a84fea872d70
```

docker.desktop PERSONAL

Containers / my-nginx

my-nginx 76e2ad484dc nginx:latest

STATUS Running (45 minutes ago)

Logs Inspect Bind mounts Exec Files Stats

Hide file editor

Name	Note	Size	Last modified	Mode
nginx			13 days ago	drwxr-xr-x
html	VOLUME		45 minutes ago	drwxr-xr-x
50x.html	VOLUME	497 Bytes	13 days ago	-rw-r--r--
index.html	VOLUME	34 Bytes	28 seconds ago	-rw-r--r--
pam			22 days ago	drwxr-xr-x
common-account		1.1 kB	9 months ago	-rw-r--r--

/usr/share/nginx/html/index.html

HTML

```
1 <h1>Hello from Named Volume!</h1>
2
```

Terminal

```
PS C:\Users\Marawan Ragab\my_web_project> docker volume create vol-html
vol-html
PS C:\Users\Marawan Ragab\my_web_project> docker volume create vol-conf
vol-conf
PS C:\Users\Marawan Ragab\my_web_project> docker run -d --name my-nginx \
--v vol_html:/usr/share/nginx/html \
--v vol_conf:/etc/nginx/conf.d \
--nginx
76e2ad484dc9663b823eedf8eaf7bf2b338e239de76c1af1a84fea872d70
PS C:\Users\Marawan Ragab\my_web_project> docker container ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
76e2ad484dc   nginx     "/docker-entrypoint..." 44 minutes ago Up 44 minutes 80/tcp       my-nginx
PS C:\Users\Marawan Ragab\my_web_project> docker exec -it my-nginx bash -c "echo '<h1>Hello from Named Volume!</h1>' > /usr/share/nginx/html/index.html"
PS C:\Users\Marawan Ragab\my_web_project>
```

Engine running RAM 0.99 GB CPU 0.63% Disk 2.02 GB used (limit 1006.85 GB)

```
PS C:\Users\Marawan Ragab\my_web_project> docker rm -f my-nginx
my-nginx
PS C:\Users\Marawan Ragab\my_web_project>
```

docker.desktop PERSONAL

Containers Give feedback

Container CPU usage 0.00% / 800% (8 CPUs available)

Container memory usage 8.51MB / 3.68GB

Show charts

Search

Only show running containers

Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
web1	6e3663c22c54	nginx	8080:80	0%	13 seconds ago	
web2	1b7bfde0e345	nginx	8081:80	0%	1 second ago	

Walkthroughs

Multi-container applications 8 mins

Containerize your application 3 mins

View more in the Learning center

Terminal

```
PS C:\Users\Marawan Ragab\my_web_project> docker run -d --name web1 -p 8080:80 \
--v vol_html:/usr/share/nginx/html \
--v vol_conf:/etc/nginx/conf.d \
--nginx
6e3663c22c54fa9bc57eb132beb75dbc2a6b88d4c7b93a129d5a545d89bddd
PS C:\Users\Marawan Ragab\my_web_project> docker run -d --name web2 -p 8081:80 \
--v vol_html:/usr/share/nginx/html \
--v vol_conf:/etc/nginx/conf.d \
--nginx
1b7bfde0e34541637dfbd2dc3b4edf27c88e5e77d9c8628b19eb382e483c977
PS C:\Users\Marawan Ragab\my_web_project>
```

Engine running RAM 1.17 GB CPU 0.38% Disk 2.02 GB used (limit 1006.85 GB)



localhost:8080

Hello from Named Volume!



localhost:8081

Hello from Named Volume!

Problem 2:

- Run a container Nginx with name nginx-bind-mount and attach 2 volumes using bind mount under any paths
- Remove the container
- Run a new container with the following:
 - o Attach the two volumes that were attached to the previous container
 - o Check the old data in the new containers

Users > Marawan Ragab > my_web_project

Name	Date modified	Type	Size
default.conf	4/6/2026 5:49 PM	CONF File	1 KB
index.html	4/6/2026 4:06 PM	Brave HTML Docu...	1 KB

docker desktop PERSONAL

Containers Give feedback

Container CPU usage 0.00% / 800% (8 CPUs available)

Container memory usage 7.53MB / 3.68GB

Show charts

Search

Only show running containers

Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
nginx-bound-mount	1f97e9e38e5a	nginx		0%	13 seconds ago	

Walkthroughs

Multi-container applications 8 mins

Containerize your application 3 mins

View more in the Learning center

Terminal

```
PS C:\Users\Marawan Ragab\my_web_project> docker run -d --name nginx-bound-mount --v "${PWD}\index.html:/usr/share/nginx/html/index.html" --v "${PWD}\default.conf:/etc/nginx/conf.d/default.conf" --nginx 1f97e9e38e5aff9d5fa48958812dd3ca22d57dc67b18d42beb4ee66c39511c PS C:\Users\Marawan Ragab\my_web_project>
```

Engine running RAM 1.16 GB CPU 0.50% Disk: 2.02 GB used (limit 1006.85 GB)

Terminal Update available

```
PS C:\Users\Marawan Ragab\my_web_project> docker rm -f nginx-bound-mount
nginx-bound-mount
PS C:\Users\Marawan Ragab\my_web_project>
```

docker desktop PERSONAL

Containers Give feedback

Container CPU usage 0.00% / 800% (8 CPUs available)

Container memory usage 15.05MB / 3.68GB

Show charts

Search

Only show running containers

Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
web1	2c6b1628861e	nginx	8080:80	0%	1 minute ago	
web2	dca71e62fb57	nginx	8081:80	0%	1 minute ago	

Walkthroughs

Multi-container applications 8 mins

Containerize your application 3 mins

View more in the Learning center

Terminal

```
PS C:\Users\Marawan Ragab\my_web_project> docker run -d --name web1 -p 8080:80 --v "${PWD}\index.html:/usr/share/nginx/html/index.html" --v "${PWD}\default.conf:/etc/nginx/conf.d/default.conf" --nginx 2c6b1628861e033bee4e8fd85c3517a22176b4215a4dc34a64724dccc1276fa PS C:\Users\Marawan Ragab\my_web_project> docker run -d --name web2 -p 8081:80 --v "${PWD}\index.html:/usr/share/nginx/html/index.html" --v "${PWD}\default.conf:/etc/nginx/conf.d/default.conf" --nginx dca71e62fb57494c5a40b3dd4c44a78a71f79338565181c57215e48cd7e63bd3 PS C:\Users\Marawan Ragab\my_web_project>
```

Engine running RAM 1.19 GB CPU 0.00% Disk: 2.02 GB used (limit 1006.85 GB)

Terminal Update available

Hello Docker v2

Problem 3:

- Create 2 nginx containers with 2 different network of type bridge, enter to one of them and use curl command to view the content of the other container

Terminal

```
PS C:\Users\Marawan Ragab> docker network create net-alpha
e91ce23f6d1705a977b61d4fd33308c78cd96ac50bdbc0611e8c087a00ea14d2
PS C:\Users\Marawan Ragab> docker network create net-beta
4e2b83503b97655d2719be7796a8c2b729ee581721c230524a1a51346cf4fc8a
PS C:\Users\Marawan Ragab>
```

docker desktop PERSONAL

Q Search

Containers Give feedback

Container CPU usage 0.00% / 800% (8 CPUs available)

Container memory usage 11.68MB / 3.68GB

Show charts

Q Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	container-1	bec2f9067e17	nginx		0%	40 seconds ago	
☐	container-2	7ab42ffe7c3a	nginx		0%	0 seconds ago	

Walkthroughs

Multi-container applications 6 mins

Containerize your application 3 mins

View more in the Learning center

Terminal

```
PS C:\Users\Marawan Ragab> docker run -d --name container-1 --network net-alpha nginx
bec2f9067e176137c0bfc044cafb2b2e145e00cf6d90c550b643be2094a31a12
PS C:\Users\Marawan Ragab> docker run -d --name container-2 --network net-beta nginx
7ab42ffe7c3a16d31cd71d89875fadcf083baf4dd898727a787adffcdf607538
PS C:\Users\Marawan Ragab>
```

Engine running

RAM 2.35 GB CPU 0.25% Disk: 2.02 GB used (limit 1006.85 GB)

Terminal Update available

Type here to search

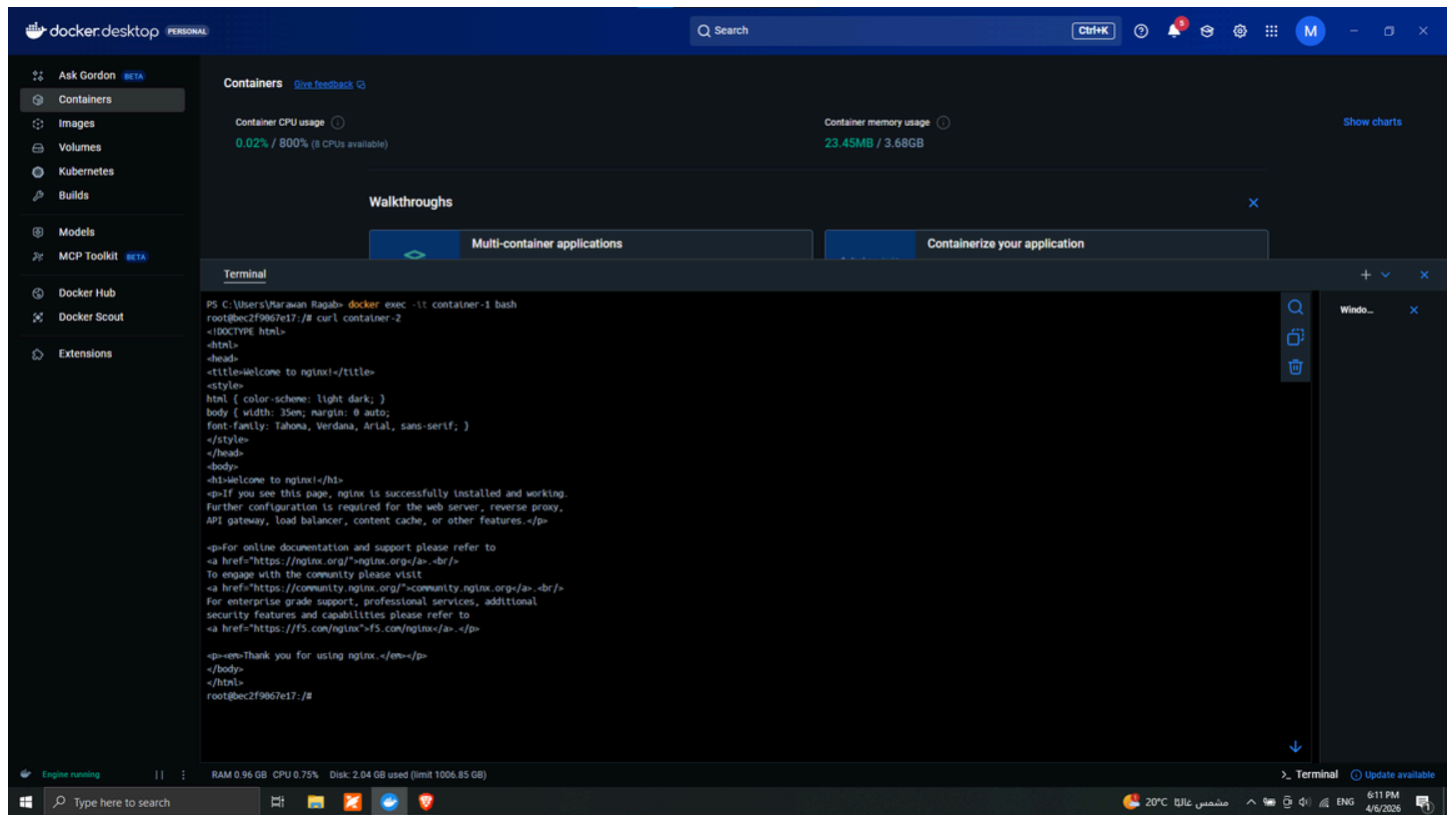
6:04 PM 4/6/2026

Terminal

```
PS C:\Users\Marawan Ragab> docker network connect net-alpha container-2
PS C:\Users\Marawan Ragab>
```

Terminal

```
PS C:\Users\Marawan Ragab> docker exec -it container-1 bash
root@bec2f9067e17:/#
```



Problem 4:

- Create docker compose with:

Two services nginx and mysql

Add needed ports and environments for both services

nginx service is depending on mysql service


```
File Edit Selection View Go Run Terminal Help
C:\Users\Marawan Ragab\my_web_project> docker-compose.yml
1 version: '3.8'
2
3 D-Run All Services
4 services:
5   D-Run Service
6   db:
7     image: mysql:8.0
8     container_name: mysql-db
9     restart: always
10    environment:
11      MYSQL_ROOT_PASSWORD: root_password
12      MYSQL_DATABASE: my_ai_db
13      MYSQL_USER: marwan_user
14      MYSQL_PASSWORD: user_password
15    ports:
16      - "3306:3306"
17   D-Run Service
18   web:
19     image: nginx:latest
20     container_name: nginx-server
21     ports:
22       - "8080:80"
23     depends_on:
24       - db
```

You have Docker installed on your system. Do you want to install the recommended 'Dev Containers' extension from Microsoft for it?

Install Show Recommendations

Ln 23, Col 13 Spaces: 4 UTF-8 CRLF Compose

19°C غائم عالما 7:23 PM 4/6/2026

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5435]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Marawan Ragab\my_web_project> docker-compose up -d
time="2026-04-06T19:22:55+02:00" level=warning msg="C:\\Users\\Marawan Ragab\\my_web_project\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 12/12
db Pulled 161.2s
0940ecf5daec Pull complete 1.7s
1378718ee973 Pull complete 0.6s
440208b3f29f Pull complete 2.1s
cd8097e7c26b Pull complete 157.0s
771773a1fe7a Pull complete 76.7s
d4c7048d1cf1 Pull complete 76.5s
5915b0e5580c Pull complete 2.3s
10ae8ef3b9fc Pull complete 153.2s
c3b8cfb10b41 Pull complete 77.0s
6375ec02106f Pull complete 76.6s
60c3210ab1f6 Pull complete 92.0s
[+] Running 2/3
Network my_web_project_default Created 0.2s
Container mysql-db Starting 6.7s
Container nginx-server Created 0.2s
Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:3306 -> 127.0.0.1:0: listen tcp 0.0.0.0:3306: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.

C:\Users\Marawan Ragab\my_web_project>
```